

MAINCRAFTS TECHNOLOGY

VLSI Design Internship: Task 2 Completion Report

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Platform Used	Icarus Verilog + GTKWave	Submission Date	August 12, 2026

1. EXECUTIVE OBJECTIVE

The core objective of Task 2 is to transition from schematic-level logic design into code-based chip design using Register Transfer Level (RTL) modeling in Verilog HDL. This report documents the implementation, testbench creation, console simulation results, and visual verification for:

- Basic Logic Gates (AND, OR, NOT, NAND, NOR, XOR)
- Half Adder Circuit
- Full Adder Circuit

2. RTL DESIGN CONCEPTS & VERILOG FUNDAMENTALS

- Verilog Module:** The structural building block in Verilog HDL that models a hardware component with defined inputs/outputs.
- Continuous Assignment (assign):** Used to model concurrent combinational logic. The expression is immediately re-evaluated and assigned to the output wire upon input changes.
- Testbench Verification:** A non-synthesizable simulation environment to instantiate the Unit Under Test (UUT), apply stimulus, and observe responses.

3. CIRCUIT IMPLEMENTATIONS, SIMULATION & RESULTS

Part A: Basic Logic Gates

1. Verilog RTL Design & Simulation Console (VS Code)

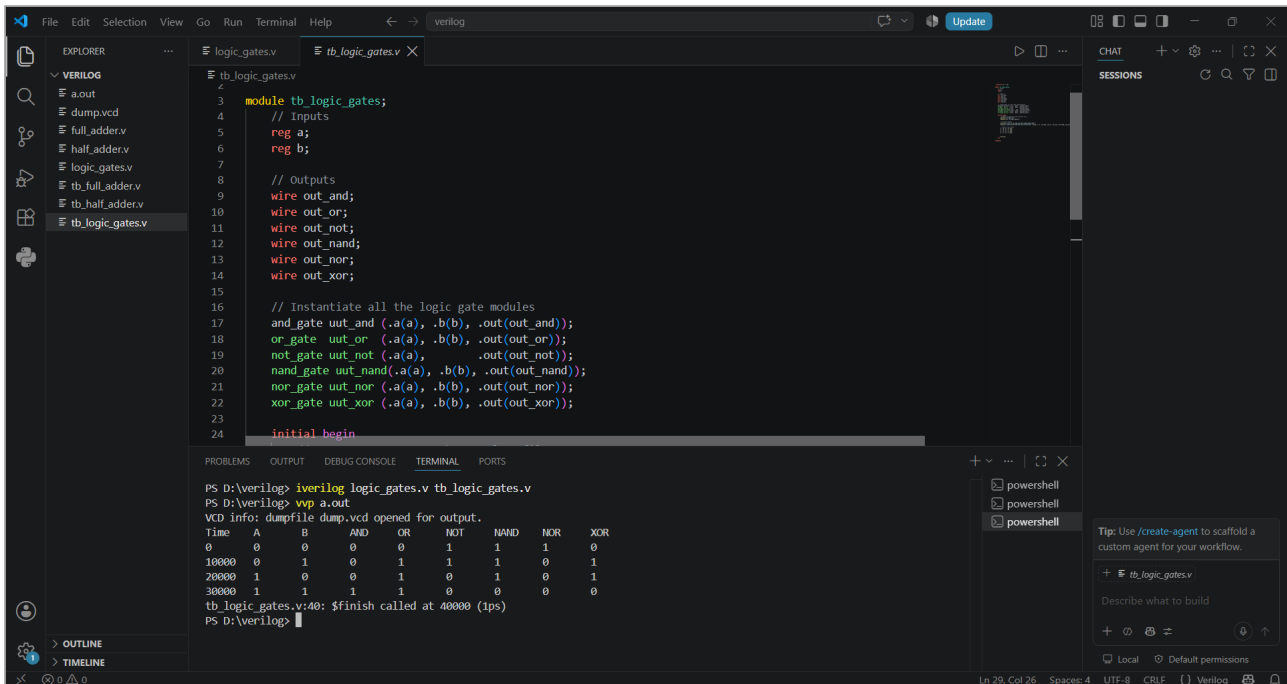


Fig 1: Module definitions, Testbench instantiation, and Terminal output for basic logic gates.

2. GTKWave Waveform Verification

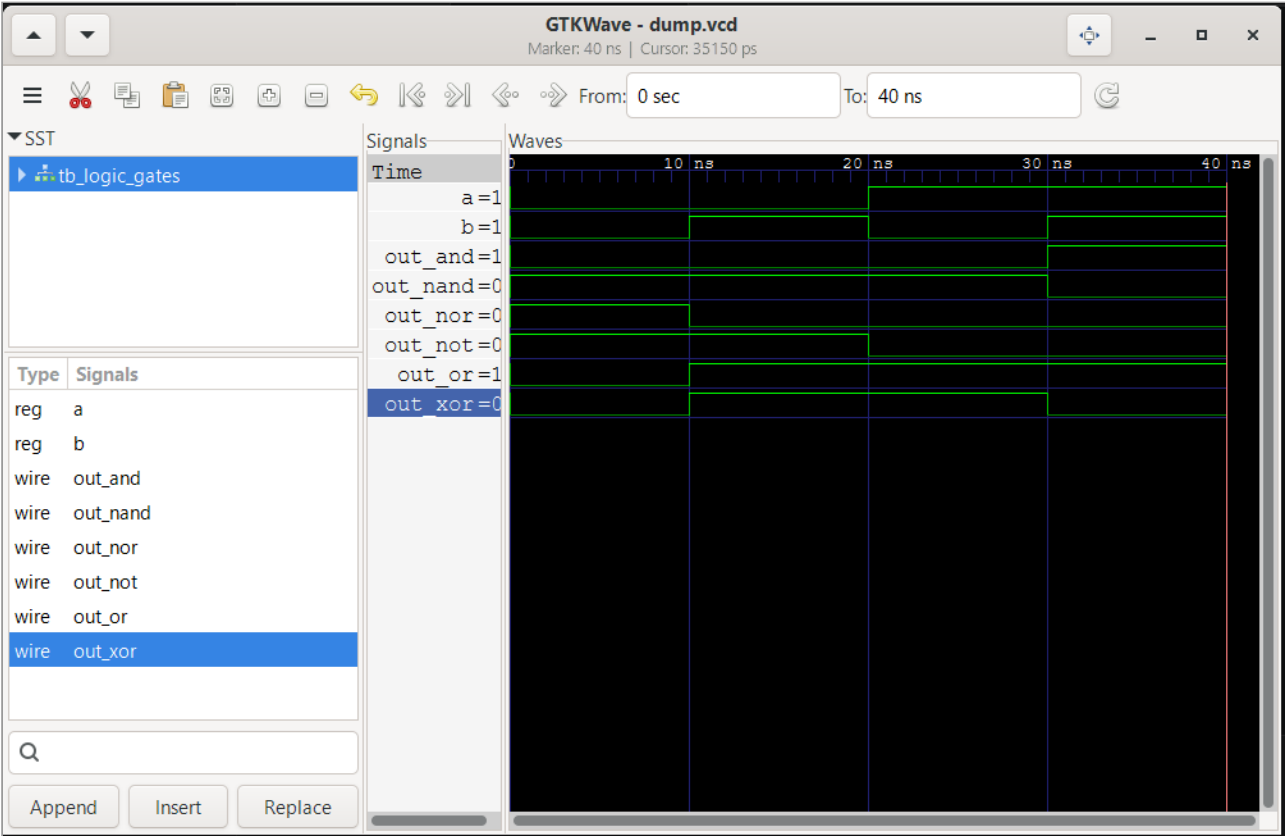


Fig 2: VCD waveform dump observed in GTKWave for basic gate permutations.

Part B: Half Adder

1. Truth Table & Logical Equations

$Sum = A \oplus B$ $Carry = A \cdot B$

A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

2. Verilog RTL Design & Simulation Console (VS Code)

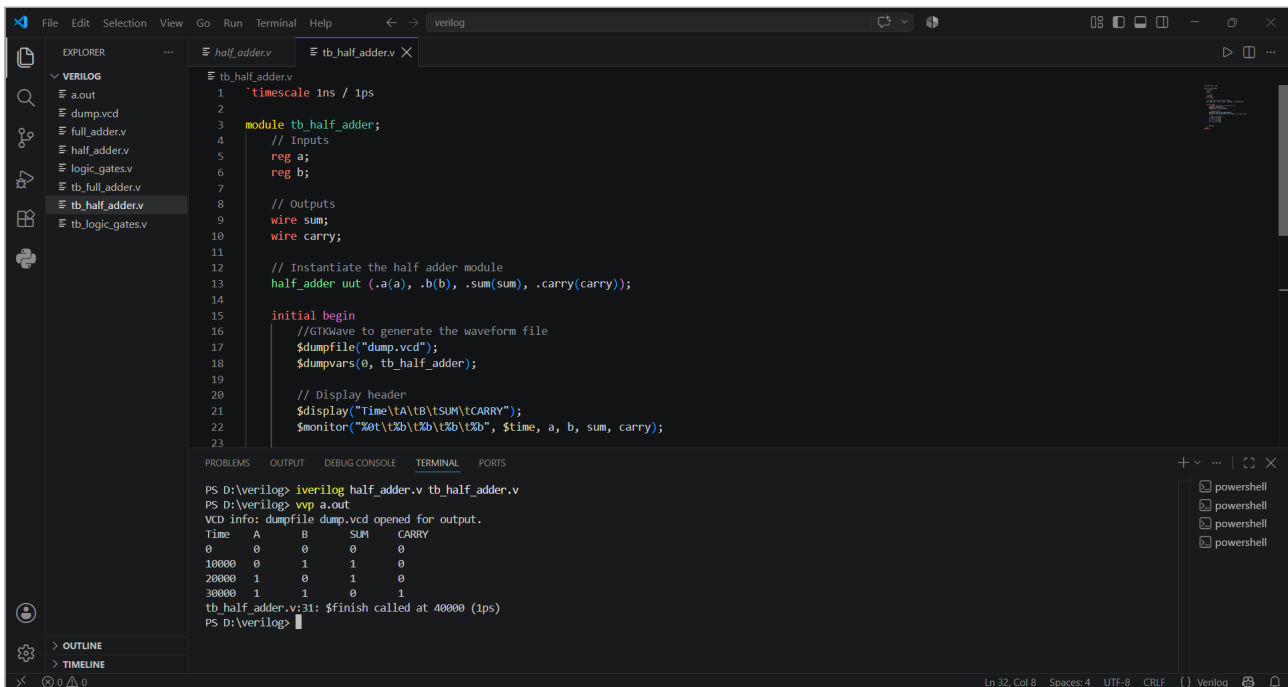


Fig 3: Half Adder continuous assignment logic and Terminal output verification.

3. GTKWave Waveform Verification

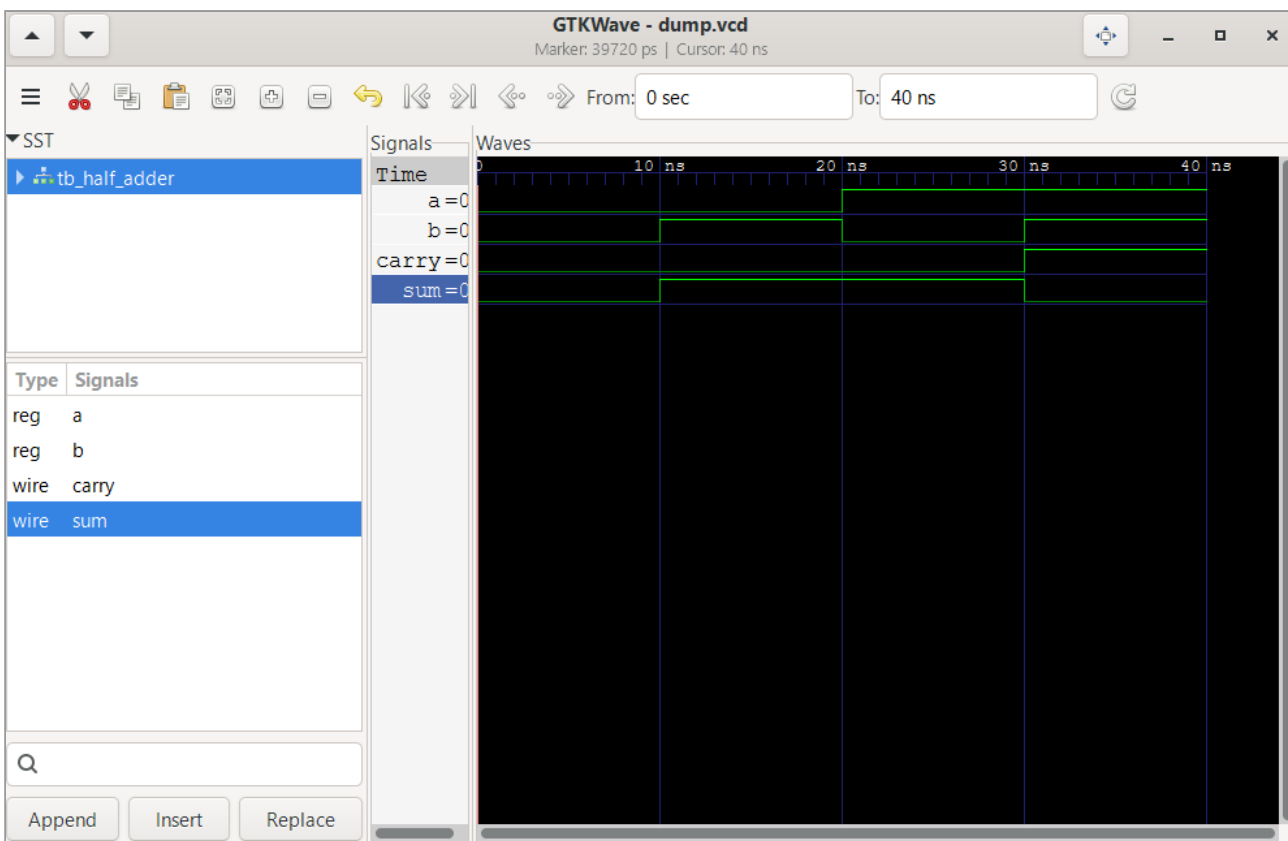


Fig 4: GTKWave tracking Carry and Sum output parity based on A and B stimuli.

Part C: Full Adder

1. Truth Table & Logical Equations

$$\text{Sum} = A \oplus B \oplus \text{Cin} \quad \text{Cout} = (A \cdot B) + (B \cdot \text{Cin}) + (A \cdot \text{Cin})$$

A	B	Cin	Sum	Cout
0	0	0	0	0

0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

2. Verilog RTL Design & Simulation Console (VS Code)

The screenshot displays the VS Code interface with a Verilog project. The Explorer panel on the left shows the file structure, including Verilog files and a dump.vcd file. The main editor shows the Verilog code for a testbench and a full adder module. The terminal at the bottom shows the simulation results, including the VCD file path and the output of the \$monitor command.

```

module tb_full_adder;
    reg b;
    reg cin;
    // Outputs
    wire sum;
    wire cout;

    // Instantiate the full adder module
    full_adder uut (.a(a), .b(b), .cin(cin), .sum(sum), .cout(cout));

    initial begin
        //GTKWave to generate the waveform file
        $dumpfile("dump.vcd");
        $dumpvars(0, tb_full_adder);

        // Display header
        $display("Time\tA\tB\tCIN\tSUM\tCOUT");
        $monitor("%0t\t%b\t%b\t%b\t%b\t%b", $time, a, b, cin, sum, cout);

        // Apply truth table
        a = 0; b = 0; cin = 0; #10;
        a = 0; b = 0; cin = 1; #10;
    end
endmodule

```

PS D:\verilog> vvp a.out
VCD info: dumpfile dump.vcd opened for output.
Time A B CIN SUM COUT
0 0 0 0 0 0
10000 0 0 1 1 0
20000 0 1 0 1 0
30000 0 1 1 0 1
40000 1 0 0 1 0
50000 1 0 1 0 0
60000 1 1 0 0 1
70000 1 1 1 1 1
tb_full_adder.v:34: \$finish called at 80000 (1ps)
PS D:\verilog>

Fig 5: Full Adder RTL integration with 3-bit input combinations.

3. GTKWave Waveform Verification

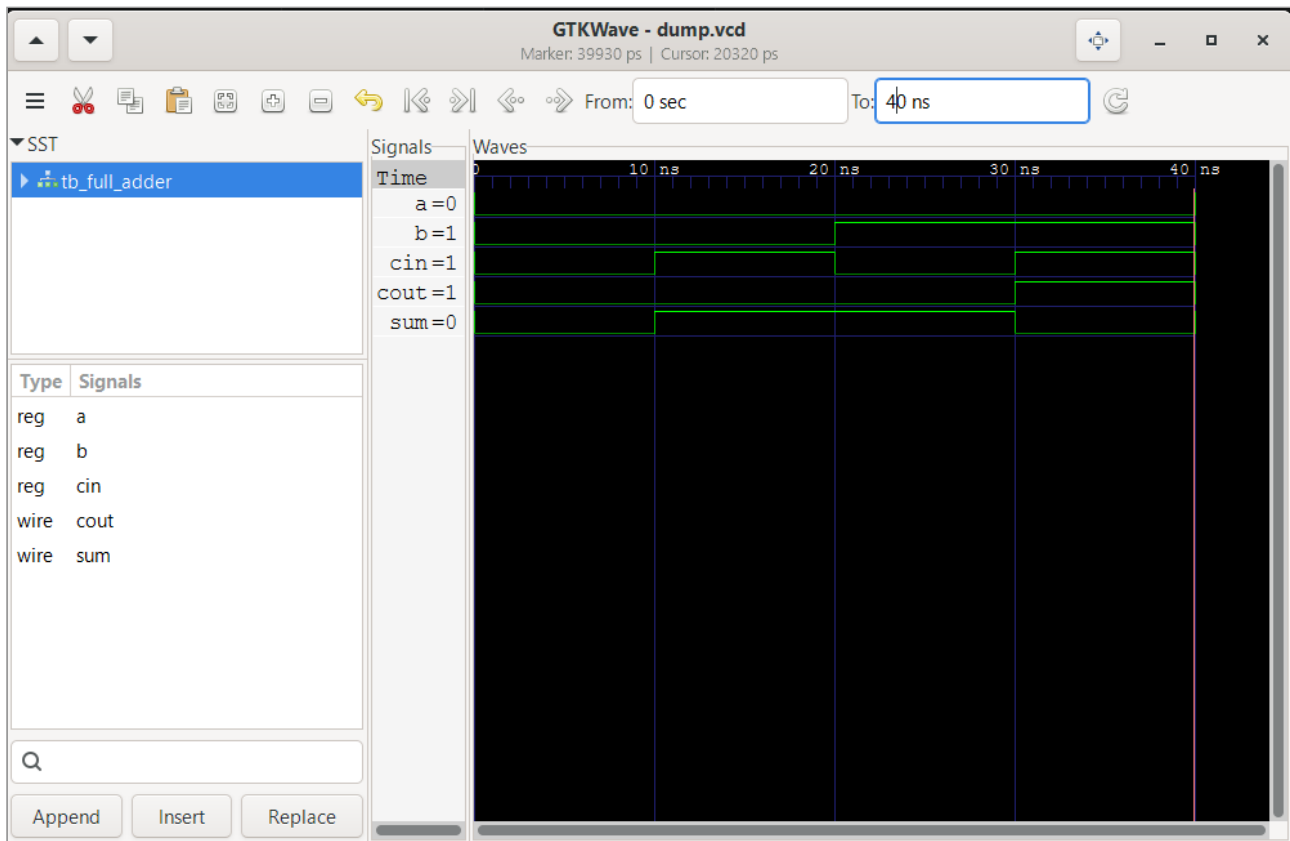


Fig 6: VCD output reflecting correct Carry-Out propagation across all 8 states.

4. KEY TAKEAWAYS & CONCLUSION

- **Functional Correctness:** All continuous assignment statements successfully modeled combinational gate logic and arithmetic addition, eliminating unintended latches or high-impedance states.
- **Simulation Workflow:** Utilized \$dumpfile and \$dumpvars to generate VCD files for real-time visual inspection in GTKWave.
- **Task Completion:** RTL modules, testbenches, console logs, and visual waveform verifications successfully satisfy all mandatory deliverables for Task 2.